

Simple Probability

How likely is it?

$$P(\text{event}) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

Example: Bag of Marbles



A bag has 3 red marbles and 2 blue marbles (5 total).

$$P(\text{red}) = \frac{3}{5} = 0.6$$

Likelihood Scale (0 to 1)



Values closer to 0 are less likely.

Values closer to 1 are more likely.

Two Ways to Find Probability

1

Experimental Probability

- ✓ Based on results from trials or an experiment.
- ✓ The more trials, the more reliable the estimate.

2

Theoretical Probability

- ✓ Based on what is possible.
- ✓ Use the formula:
 $P(\text{event}) = \text{favorable outcomes} \div \text{total outcomes}$



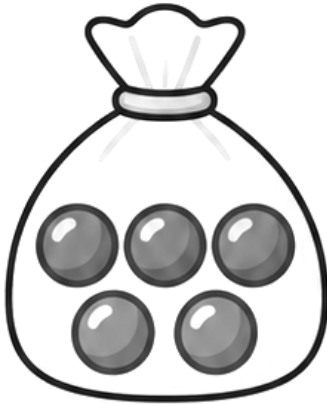
Probability is always between 0 and 1.

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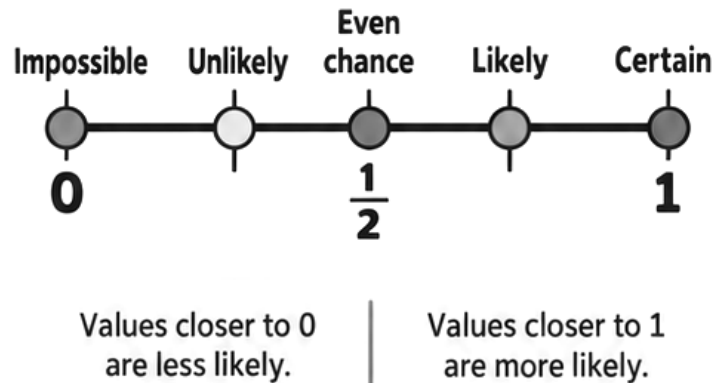
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